Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/17/2010
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Novel Electro-hydrodynamic Technology for Gas-Liquid Phase Separation in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	khusid@njit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	973-596-3316
Organization Name:	New Jersey Institute of Technology		
PI Address 1:	Chemical & Materials Engineering		
PI Address 2:	University Heights		
PI Web Page:			
City:	Newark	**State:**	NJ
Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	05/19/2009	**End Date:**	05/18/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
Contact Email:	Arnon.Chait@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	James, Conrad (Sandia National Laboratories)		
Grant/Contract No.:	NNX09AK06G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In space, the lack of the gravity-driven gas-liquid phase separation has severely compromised a wide range of space technologies, ranging from liquid pumps to volumetric techniques for sampling liquids and to electrolytic oxygen generators. Unexpectedly, while carrying out experiments on suspensions in microgravity aboard the NASA aircraft, we observed the appearance of a gas-liquid phase separation caused by the application of a high-gradient electric fields The mechanism of this separation is based on polarization forces generated by spatially nonuniform electric fields in a gas-liquid mixture due to the dielectric constant and conductivity mismatch between the gas phase and the liquid phase. The project focuses on the development of engineering fundamentals of a technology utilizing high-gradient electric fields for the phase separation of gas-liquid systems in microgravity. The main outcome of the proposed highly integrated theoretical studies, ground-based experiments, and short-term tests in reduced gravity aboard the NASA aircraft will include an experimentally validated mathematical model for predicting electro-hydrodynamic phenomena in gas-liquid mixtures and the scalable design of devices for a gas-liquid phase separation in advanced thermal and fluid		

	management systems for space applications. The research will be integrated with educational and outreach initiatives, including graduate and undergraduate training in space sciences and related technology areas with emphasis on the recruitment of women and minorities. High school students and teachers will be given the opportunity for research experience through the PI involvement in the NASA-sponsored educational programs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research effort improves our understanding of the behavior of gas-liquid mixtures subject to a high-gradient electric field and facilitate the design and optimization of devices to mitigate these undesirable phenomena in microgravity. The method developed for stripping bubbles from a droplet offers the potential for an in-line debubbler in a pre-processing stage of lab-on-a-chip (LOC) systems for bodily fluids.
Task Progress:	The most straightforward application of the proposed field and flow driven bubble stripping and/or two phase flow management is in lab-on-a-chip (LOC) devices now envisioned for space applications. Such devices inherently have air bubbles which are difficult to remove. Existing methods of the bubble removal in LOC systems appear to be of limited use for space applications due to different functional requirements. A new method for the field and flow driven concentration and removal of bubbles from droplets of aqueous-based polymer solutions that we have developed overcomes the critical challenges facing current technologies for gas-liquid separation and appears to meet the stringent requirements of LOC devices for testing bodily fluids, such as saliva and urine, in microgravity. To eliminate adverse effects of electro-chemical reactions at the fluid-electrode interface, a fluid droplet is placed onto electrically insulating films and serves as a floating electrode capacitively coupled to external electrodes. Following the application of alternating current voltage, bubbles accumulate on the drop surface, forming a layer of dense foam possessing yield stress. A short voltage pulse is then applied to transfer part of drop enriched with bubbles to a ground electrode. Experiments were conducted within the range of parameters appropriate for future applications of LOC devices in microgravity. The research is integrated with educational and outreach initiatives in the areas of gas-liquid flow, electro-hydrodynamics, and microfluidics. Two NJIT graduate students, one PhD (minority) and one MS, are involved in the ongoing research. An undergraduate student, a high school student, and a high school teacher will be working in the PI's lab on this project in the summer of 2010 within the New York City Research Initiative (NYCRI) program sponsored by the NASA Goddard Institute for Space Studies.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Articles in Peer-reviewed Journals	Lo TS, Khusid B, Koplik J. "Field induced alignment of flexible polyelectrolytes in solution." Physical Review Letters, in press, March 2010. , Mar-2010